Languages, automata and computation II
Tutorial 2

Winter semester 2023/2024

In this tutorial we explore weighted automata and linear recursive sequences over a field. In particular, we will be concerned with functions in $\Sigma^* \to \mathbb{Q}$ for a finite alphabet Σ.

Exercise 1. Show that the set of functions $\Sigma^* \to \mathbb{Q}$ can be given the structure of a vector space over $\mathbb{Q}$. What is its dimension?

Solution: Define the addition of functions as $(f + g)(w) := f(w) + g(w)$ and scalar multiplication by $\alpha \in \mathbb{Q}$ as $(\alpha \cdot f)(w) := \alpha \cdot f(w)$. It can be checked that these definitions satisfy the requirements of vector spaces.

The space is infinite dimensional: Functions f_w's s.t.$f_w(w) = 1$ and $f_w(u) = 0$ if $u \neq w$ are linearly independent. □

Rational functions and their linear representations

A *linear representation* over Σ is a triple $A = (x, M, y)$ where the transition matrix $M : \Sigma \to \mathbb{Q}^{k \times k}$ maps each letter $a \in \Sigma$ to a $k \times k$ rational matrix M_a, $x : \mathbb{Q}^{1 \times k}$ is a row vector, and $y : \mathbb{Q}^{k \times 1}$ is a column vector. The transition matrix M is extended homomorphically to a function $\Sigma^* \to \mathbb{Q}^{k \times k}$ (where matrices form a ring with the usual notions of matrix sum and product). The semantics of a linear representation is the function $f : \Sigma^* \to \mathbb{Q}$ s.t.

$$f(w) = x \cdot M(w) \cdot y, \quad \text{for every } w \in \Sigma^*.$$

Call a function *rational* if it is of the form above.

Exercise 2 (name of the game)**.** Consider the special case of a unary alphabet $\Sigma = \{a\}$. Define the *generating series* of $f : \mathbb{N} \to \mathbb{Q}$ to be

$$f(x) = \sum_{n=0}^{\infty} f(n) \cdot x^n$$

Show that if f is rational iff its generating series $f(x)$ is a rational power series. Recall that rational power series are those which can be written as $p(x)/q(x)$ for two polynomials $p, q \in \mathbb{Q}[x]$.

Solution: For the "only if" direction, assume f is rational and thus $f(n) = u \cdot M^n \cdot v$. Then its generating series satisfies

$$\begin{aligned} f(x) &= \sum_n (u \cdot M^n \cdot v) x^n = \\ &= u \cdot \sum_n (Mx)^n \cdot v = \\ &= u \cdot (I - Mx)^{-1} \cdot v, \end{aligned}$$

where one observes that $I - Mx$ is invertible (even over the ring of power series matrices) and that the inverse of a polynomial matrix is a rational matrix.

For the "if " direction, let $f(x) = \frac{p(x)}{1 - x \cdot q(x)}$. Thus $f(x) = p(x) + x \cdot q(x) \cdot f(x)$ and one notices that this induces a linear recurrence for $f(n)$. □

A matrix $A \in \mathbb{Q}^{k \times k}$ is *deterministic* if each row has at most one nonzero entry. A rational function f with linear representation (x, M, y) is *deterministic* if M_a, M_b are deterministic and x has at most one nonzero entry.

Exercise 3. Show that there are rational functions which are not deterministic.

Solution: A deterministic rational function never uses the "+" operation. In other words, if f is deterministic then $f(w)$ is a *product* of numbers appearing in x, M, and y. In particular, the image of f cannot contain numbers with arbitrarily large prime divisors. Now consider the function $f(w) = |w|$. It is rational and its image is $\mathbb{N}$. Since there are infinitely many primes, f cannot be deterministic. □

q-finite functions

Given a function $f : \Sigma^* \to \mathbb{Q}$ and a word $u \in \Sigma^*$, let the *left quotient* $u^{-1}f : \Sigma^* \to \mathbb{Q}$ be the function defined as

$$(u^{-1}f)(w) = f(uw), \quad \text{for every } w \in \Sigma^*.$$

Call a function f *q-finite* if the set of left quotients

$$\left\{ u^{-1}f \mid u \in \Sigma^* \right\}$$

span a finite-dimensional subspace of $\Sigma^* \to \mathbb{Q}$.

Exercise 4. Show that f is q-finite iff it is rational.

Solution: For the "only if" direction, assume that f is q-finite. There is a dimension d and this many basis left quotients

$$f_1 := u_1^{-1}f, \ldots, f_d := u_d^{-1}f$$

s.t. every left quotient is a linear combination of $f_1, \ldots, f_d$. We now construct a linear representation for f. Consider a basis left quotient f_m and extend it by reading $a \in \Sigma$ to the quotient $(u_m a)^{-1}f$. This quotient is not necessarily a basis element, however it can be written as a (unique) linear combination of basis elements

$$(u_m a)^{-1}f = \alpha_{m,1} \cdot f_1 + \cdots + \alpha_{m,d} \cdot f_d.$$

This defines $M_a(m,n) := \alpha_{m,n}$. To define the initial row vector we write f itself as

$$f = \alpha_1 \cdot f_1 + \cdots + \alpha_d \cdot f_d,$$

giving $x = (\alpha_1, \ldots, \alpha_d)$, and the final column vector is obtained by evaluating the basis elements at ε, giving $y = (f_1(\varepsilon), \ldots, f_d(\varepsilon))^T$.

Correctness amounts to prove

$$f(w) = x \cdot M(w) \cdot y, \quad \text{for all } w \in \Sigma^*.$$

This will follow at once from the following inductive property:

$$w^{-1}f = x \cdot M(w) \cdot (f_1, \ldots, f_d)^T, \quad \text{for all } w \in \Sigma^*.$$

For $w = \varepsilon$ it holds by the definition of x. Inductively we have

$$\begin{aligned}
(wa)^{-1}f = a^{-1}w^{-1}f = \\
&= a^{-1}(x \cdot M(w) \cdot (f_1, \ldots, f_d)^T) = \\
&= x \cdot M(w) \cdot \left(a^{-1}f_1, \ldots, a^{-1}f_d\right)^T = \\
&= x \cdot M(w) \cdot (M_a \cdot (f_1, \ldots, f_d)^T) = \\
&= x \cdot M_{wa} \cdot (f_1, \ldots, f_d)^T.
\end{aligned}$$

We have used the inductive assumption, the fact that left quotients act linearly, and the definition of M_a.

For the "if" direction, assume that f is rational. There is a k-dimensional linear representation (x, M, y) s.t. $f(w) = x \cdot M(w) \cdot y$ for every $w \in \Sigma^*$. The set of $k \times k$ matrices $\mathbb{Q}^{k \times k}$ is a vector space of dimension k^2 (with respect to matrix addition and scalar multiplication). Now consider the linear span of all reachable matrices

$$V := \mathsf{span}(M(w) \mid w \in \Sigma^*) \subseteq \mathbb{Q}^{k \times k}.$$

As a subspace of a vector space of dimension k^2 it is itself of some finite dimension $d \leq k^2$. Let a basis of V be $M_1 := M_{u_1}, \ldots, M_d := M_{u_d}$. We claim that $f_1, \ldots, f_d$ is a basis of the vector subspace generated by left quotients of f, where for every $1 \leq i \leq d$,

$$f_i(w) := x \cdot M_{u_i \cdot w} \cdot y, \quad \text{for all } w \in \Sigma^*.$$

First of all, f_i is indeed a left quotient of f. Secondly, let $u^{-1}f$ be a left quotient for f. Since $M(u)$ is in V, by the spanning property of the basis we can write

$$M(u) = \alpha_1 \cdot M_1 + \cdots + \alpha_d \cdot M_d.$$

For every input word $w \in \Sigma^*$ we can write

$$\begin{aligned}
(u^{-1}f)(w) = f(uw) = x \cdot M(uw) \cdot y = \\
&= x \cdot M(u) \cdot M(w) \cdot y = \\
&= x \cdot (\alpha_1 \cdot M_1 + \cdots + \alpha_d \cdot M_d) \cdot M(w) \cdot y = \\
&= \alpha_1 \cdot x \cdot M(u_1 \cdot w) \cdot y + \cdots + \alpha_d \cdot x \cdot M(u_d \cdot w) \cdot y = \\
&= \alpha_1 \cdot f(u_1 \cdot w) + \cdots + \alpha_d \cdot f(u_d \cdot w) = \\
&= \alpha_1 \cdot (u_1^{-1}f)(w) + \cdots + \alpha_d \cdot (u_d^{-1}f)(w) = \\
&= \alpha_1 \cdot f_1(w) + \cdots + \alpha_d \cdot f_d(w).
\end{aligned}$$

Since w was arbitrary, we have established $u^{-1}f = \alpha_1 \cdot f_1 + \cdots + \alpha_d \cdot f_d$, as required. □

Closure properties

Exercise 5. Show that the set of all q-finite functions is a vector subspace of $\Sigma^* \to \mathbb{Q}$.

Solution: This boils down to showing that q-finite functions are closed under multiplication by constants and by addition. Both verifications are immediate by applying linearity either to linear representations or to left quotients. (Left quotienting acts linearly: $u^{-1}(\alpha \cdot f) = \alpha \cdot (u^{-1}f)$ and $u^{-1}(f+g) = u^{-1}f + u^{-1}g$). □

Exercise 6. Show that the set of q-finite functions is closed under the following operations.

1. Hadamard product: $(f \cdot g)(w) := f(w) \cdot g(w)$.
2. Cauchy product: $(f * g)(w) := \sum_{uv=w} f(u) \cdot g(v)$.
3. Iteration, when $f(\varepsilon) = 0$: $f^* := f^0 + f^1 + f^2 + \cdots$, where $f^0(w)$ is 1 if $w = \varepsilon$ and 0 otherwise, and $f^{n+1} = f^n * f$ for every $n \geq 0$.

Solution:

1. Let $f_1, \ldots, f_d$ be a basis for f and $g_1, \ldots, g_e$ one for g. We claim that quotients of $f \cdot g$ are in the linear span of

$$\{f_i \cdot g_j \mid 1 \leq i \leq d, 1 \leq j \leq e\}.$$

   This follows at once since (a) quotients distribute over Hadamard product, and (b) Hadamard product is bilinear. This shows that the dimension of $f \cdot g$ is at most $d \cdot e$, but it could be less.

2. For words $u = a_1 \cdots a_n, w \in \Sigma^*$, by analysing the Cauchy product we have that $u^{-1}(f * g)$ equals

$$\underbrace{f(\varepsilon)}_{\in\mathbb{Q}} \cdot u^{-1}g + \underbrace{f(a_1)}_{\in\mathbb{Q}} \cdot (a_2 \cdots a_n)^{-1}g + \cdots + \underbrace{f(a_1 \cdots a_{n-1})}_{\in\mathbb{Q}} \cdot a_n^{-1}g + u^{-1}f * g.$$

   In other words, left quotients of $f * g$ are linear combinations of left quotients of g and $h * g$ with h a left quotient of f. This suggests that if $f_1, \ldots, f_d$ is a basis for f and $g_1, \ldots, g_e$ one for g, then left quotients of $f * g$ are in the linear span of

$$\{f_i * g \mid 1 \leq i \leq d\} \cup \{g_j \mid 1 \leq j \leq e\}.$$

   This can be verified thanks to the calculation above and bilinearity of Cauchy product. Incidentally, this shows that the dimension of $f * g$ is at most $d + e$.

3. First notice that iteration is well-defined, thanks to the condition $f(\varepsilon) = 0$. Let $f_1, \ldots, f_d$ be a basis for left quotients of f and let $f_0 := f^0$. We claim that left quotients of f^* are in the linear span of

$$\{f_0 * f^*, f_1 * f^*, \ldots, f_d * f^*\}.$$

We show that every left quotient $u^{-1}f^*$ is a linear combination of elements above by induction on the length of u. In the base case, $u = \varepsilon$ and thus $\varepsilon^{-1}f^* = f^*$ is already the basis element $f_0 * f^*$.

In the inductive case, let $u = a_1 \cdots a_n$ have positive length $n \geq 1$. We use again $f^* = f^0 + f * f^*$ to write

$$u^{-1}f^* = u^{-1}(f^0 + f * f^*) = \underbrace{u^{-1}f^0}_{0} + u^{-1}(f * f^*) =$$

$$= \underbrace{f(\varepsilon) \cdot u^{-1}f^*}_{0} + f(a_1) \cdot (a_2 \cdots a_n)^{-1}f^* + \cdots + f(a_1a_2 \cdots a_{n-1}) \cdot a_n^{-1}f^* + u^{-1}f * f^*.$$

This shows that $u^{-1}f^*$ is a linear combination of shorter quotients of f^* and $u^{-1}f * f^*$. By the inductive assumption shorter quotients of f^* are in the span of the perspective basis. Regarding $u^{-1}f * f^*$, since $u^{-1}f$ is a linear combination of $f_1, \ldots, f_d$, by left linearity $u^{-1}f * f^*$ is a linear combination of $f_1 * f^*, \ldots, f_d * f^*$.

□

Exercise 7 (Inverses)**.** 1. Under which condition does f have an inverse w.r.t. Hadamard product? Is Hadamard-invertibility decidable?

2. Under which condition does f have an inverse w.r.t. Cauchy product?

3. In the latter case, find an expression for the Cauchy inverse of f.

Solution: 1. First of all the unit for the Hadamard product is the constantly 1 function. The function f has a Hadamard inverse iff $f(w) \neq 0$ for all w, in which case the Hadamard inverse of f is g defined as $g(w) = 1/f(w)$ for every $w \in \Sigma^*$. Hadamard invertibility is undecidable, since the complement problem (is there some w s.t. $f(w) = 0$) is undecidable for weighted automata.

2. The unit for the Cauchy product is the function f s.t. $f(\varepsilon) = 1$ and that is zero everywhere else. Call this function $\mathbb{1}$ for convenience. In order for $f * g$ to be the Cauchy unit, it is necessary that $f(\varepsilon) \neq 0$. In fact this condition is also equivalent to the existence of a Cauchy inverse, which we will compute in the next point.

3. First we show how to invert f s.t. $f(\varepsilon) = 1$. Let $g = \mathbb{1} - f$ and we claim that g^* is the Cauchy inverse of f. Indeed g^* is well defined since $g(\varepsilon) = 0$, and we have

$$f * g^* = (\mathbb{1} - g) * (g^0 + g^1 + \cdots) = g^0 = \mathbb{1}.$$ □

Regular expressions

Exercise 8 (Kleene-Schützenberger theorem)**.** Call a function *regular* if it can be generated by the following abstract grammar

$$f, g ::= p \mid \alpha \cdot f \mid f + g \mid f * g \mid f^*,$$

where p is a polynomial (function with finite support) and iteration f^* is only applied when defined. Show that a function is regular iff it is rational.

Solution: The "only if" direction follows by the closure properties of rational functions. The converse direction is more involved and we give only the proof idea. Let f be rational, thus there is a linear representation $(x, \{M_a, M_b\}, y)$, with $M_a, M_b \in \mathbb{Q}^{k\times k}$, s.t. $f(w) = x \cdot M(w) \cdot y$. Here M is in $\Sigma^* \to \mathbb{Q}^{k\times k}$. The latter set is naturally endowed by a Kleene algebra structure by the operations of sum, product, iteration (when defined), and constants 0 and 1:

$$\mathbb{S} = \left(\Sigma^* \to \mathbb{Q}^{k\times k}, +_{\mathbb{S}}, \cdot_{\mathbb{S}}, (_)^*_{\mathbb{S}}, 0_{\mathbb{S}}, 1_{\mathbb{S}}\right).$$

It is fruitful to notice that the latter is isomorphic to the matrix Kleene algebra

$$\mathbb{M} = \left((\Sigma^* \to \mathbb{Q})^{k\times k}, +_{\mathbb{M}}, \cdot_{\mathbb{M}}, (_)^*_{\mathbb{M}}, 0_{\mathbb{M}}, 1_{\mathbb{M}}\right).$$

where the definitions of sum, product, and 0, 1 are automatically inherited from the base ring $\Sigma^* \to \mathbb{Q}$; iteration is defined as $M^* := \sum_n M^n$ (when it exists). Indeed, we can map a function $M \in \mathbb{S}$ to the matrix $\widetilde{M} \in \mathbb{M}$ s.t. $\widetilde{M}_{ij}(w) := M(w)_{ij}$.

The *support* of a matrix $M \in \mathbb{M}$ is the union of the supports of all its entries. Call a matrix $M \in \mathbb{M}$ *regular* if it can be finitely generated from matrices of finite support by the algebra operations. For instance, if $M \in \mathbb{S}$ is generated by matrices $M_a, M_b \in \mathbb{Q}^{k\times k}$ in the sense of linear representations, then $\widetilde{M} \in \mathbb{M}$ is regular since

$$\widetilde{M} = (A + B)^*,$$

where $A, B \in \mathbb{M}$ have finite support and are defined as follows: A_{ij} maps a to the i, j component of M_a, and maps any other word to zero; similarly for B.

Then one shows that if $M, N \in \mathbb{M}$ are two matrices with all entries regular (in the sense of $\mathbb{S}$), then the same holds for $M + N$, $M \cdot N$, and M^*. Only the last case is non-trivial, but it can be shown by induction on the dimension of M by a suitable rule expressing M^* in terms of iteration, sum, and product of submatrices. □

Supports

The *support* of a function $f : \Sigma^* \to \mathbb{Q}$, denoted $\mathsf{supp}\, f$, is the set of words where f is nonzero. A *rational support* is the support of a rational function. Since we do not consider any other kind of support, we just say "support" for "rational support" in the following.

Exercise 9. 1. Show that the class of supports includes all regular languages.

2. Are there nonregular supports?

Solution: For a language $L \subseteq \Sigma^*$, we can define its *characteristic function* $f_L : \Sigma^* \to \mathbb{Q}$ by mapping words in the language to 1, and the rest to 0. Clearly the support of f_L is L. We now argue that if L is regular, then f_L is a rational function. It will be convenient to use regular expressions and notice how

characteristic functions interact with rational operations on languages:

$$\begin{aligned}
f_{\{\varepsilon\}} &= \mathbb{1} \\
f_{L\cap M} &= f_L \cdot f_M && \text{(Hadamard product)} \\
f_{\Sigma^*\setminus L} &= 1 - f_L && \text{(where 1 is one everywhere)} \\
f_{L\cup M} &= f_L + f_M - f_L \cdot f_M \\
f_{L\cdot M} &= f_L * f_M && \text{(Cauchy product)} \\
f_{L^*} &= f_L^*. && \text{(if } \varepsilon \notin L\text{)}
\end{aligned}$$

Finite languages are clearly supports (of polynomials). The proof is concluded by writing L as a regular expression e and applying the rules above by structural induction on e, using the closure properties of rational functions.

There are nonregular supports. We show a rational function f whose co-support is $L = \{a^n b^n \mid n \in \mathbb{N}\}$ (the complement of which is nonregular). Let $f = (g-h)^2 + \ell$ (Hadamard square), where $g(a^m b^n) = 2^m 3^n$ (and zero otherwise), $h(a^m b^n) = 2^n 3^m$ (and zero otherwise), and ℓ is the characteristic function of $\Sigma^* \setminus a^*b^*$. We have $f(w) = 0$ if $w = a^n b^n$ and f is nonzero otherwise. □

Exercise 10 (Weak cancellation property [1])**.** A language $L \subseteq \Sigma^*$ has the *weak cancellation property* if there exists a $n \in \mathbb{N}$ s.t. no matter how we split a word $w \in L$ as $w = xy_1 \cdots y_n z$ with $y_1, \ldots, y_n$ nonempty, we can always find $i, j \in \mathbb{N}$ s.t. $1 \le i \le j \le n$ and $xy_1 \cdots y_{i-1} y_{j+1} \cdots y_n z \in L$.

1. Show that supports have the weak cancellation property.
2. Find a context-free language which is not a support.

Solution: 1. Let $L = \mathsf{supp}\, f$ be the support of a rational function f with linear representation (u, M, v) of dimension k. We show that L has the weak cancellation property for $n := k$. Consider $w = xy_1 \cdots y_k z \in L$. Thus $uM(w)v \neq 0$, and in particular row vectors in the following sequence are nonzero:

$$uM(x), uM(xy_1), \ldots, uM(xy_1 \cdots y_k) \in \mathbb{Q}^{1\times k}.$$

Since there are $k+1$ vectors in the sequence above and they lie in a k-dimensional vector space, there is $1 \le j \le k$ s.t. the j-th vector is a linear combination of preceding vectors,

$$uM(xy_1 \cdots y_j) = \sum_{0\le i<j} \alpha_i \cdot uM(xy_1 \cdots y_i).$$

We now right multiply both sides by $M(y_{j+1} \cdots y_k)v$ and obtain

$$uM(w)v = \sum_{0\le i<j} \alpha_i \cdot uM(xy_1 \cdots y_i y_{j+1} \cdots y_k)v.$$

Since $w \in L$, the r.h.s. is nonzero, thus there is $0 \le i < j$ s.t.

$$uM(xy_1 \cdots y_i y_{j+1} \cdots y_k)v \neq 0.$$

This means $xy_1 \cdots y_i y_{j+1} \cdots y_k \in L$, as required.

2. Consider the context-free language $L = \{a^n b^n \mid n \in \mathbb{N}\}$. We show that it does not satisfy the weak cancellation property. For $n \in \mathbb{N}$ consider $w = a^n b^n$ split as $w_n = xy_1 \cdots y_n z$ where $x = \varepsilon$, $y_1 = \cdots = y_n = a$, and $z = b^n$. Clearly there is no way to remove any infix $u_i \cdots u_j$ of a^n from w and remain in L. □

Exercise 11. Are the following problems decidable for supports:

1. emptiness?
2. universality?
3. equivalence?
4. inclusion?

Solution: Emptiness is the same as non-zeroness, thus it is decidable. Non-universality on the other hand asks whether some word has zero semantics, which is undecidable. Equivalence and inclusion are more general than universality, so also undecidable. □

Exercise 12. Are supports closed under

1. intersection?
2. union?
3. concatenation?
4. Kleene star?
5. complement?

Solution: The closure properties for the first four points follow by the equations (since rational functions are closed under the respective operations)

$$\begin{aligned}
\mathsf{supp}\, f \cap \mathsf{supp}\, g &= \mathsf{supp}\,(f \cdot g), && \text{(Hadamard product)}\\
\mathsf{supp}\, f \cup \mathsf{supp}\, g &= \mathsf{supp}\,(f \cdot f + g \cdot g), && \text{(Hadamard square)}\\
\mathsf{supp}\, f \cdot \mathsf{supp}\, g &= \mathsf{supp}\,(f * g), && \text{(Cauchy product)}\\
\mathsf{supp}\, f^* &= \mathsf{supp}\, f^*. && \text{(Cauchy iteration)}
\end{aligned}$$

Supports are not closed under complement: We have seen that $a^n b^n$ is not a support, however its complement is. □

References

[1] Antonio Restivo and Christophe Reutenauer. On cancellation properties of languages which are supports of rational power series. *J. Comput. Syst. Sci.*, 29(2):153–159, October 1984.